

Hamish Chhagan

Full Stack Developer

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Projects

AI Content Generation & WordPress Publishing Platform

- Architected and developed a production-grade SaaS platform combining multi-agent AI systems, WordPress automation, and SEO optimisation using LangChain, OpenAI GPT-4, and LangGraph for complex workflow orchestration.
- Built 9 specialized AI agent tools including Google Search Console integration, website crawling (WordPress REST API + sitemap parsing), competitor analysis, and vector-based knowledge retrieval for intelligent content generation.
- Implemented PostgreSQL with pgvector extension for semantic search across crawled website content, enabling RAG-based blog generation with automatic internal linking and brand consistency.
- Developed full-stack multi-tenant architecture with React/TypeScript frontend, Express.js backend, user authentication, bot management, and secure API key-based WordPress plugin integration.
- Created intelligent website crawler that extracts and classifies content (products, services, pages), brand information (colors, logos), and generates vector embeddings for context-aware article generation.
- Integrated OAuth 2.0 for Google Search Console API, leveraging real search performance data, content gap analysis, and trending queries to inform content strategy with 7-day caching optimisation.
- Built WordPress plugin (PHP) enabling one-click publishing with automatic category/tag creation, metadata synchronization (Yoast/Rank Math compatible), and featured image handling.

AI-Powered Lead Generation Chatbot Platform

- Built a multi-tenant chatbot platform with user authentication, bot management, and real-time chat interfaces using React, TypeScript, Node.js, Express, and PostgreSQL.
- Implemented RAG system using LangChain, FAISS vector store, and OpenAI embeddings for context-aware responses from uploaded PDF knowledge bases.
- Created dynamic tool generation system allowing chatbots to capture structured lead data based on customizable schemas, with automatic email notifications via Nodemailer.
- Migrated backend from Flask to Express.js, maintaining feature parity while improving performance and type safety with TypeScript.

Full-Stack Portfolio with CI/CD Pipeline

- Established a fully automated CI/CD pipeline using GitHub Actions for zero-downtime deployments to AWS.
- Containerised the full-stack application with Docker and Docker Compose for consistent development and production environments.

The Phone Mason - SEO & Marketing Platform

- Drove lead generation via continuous SEO improvements and targeted ad campaigns, achieving top 3 local search rankings for competitive repair keywords.
- Managed the end-to-end project lifecycle, from AWS deployment to stakeholder communication and campaign optimisation.

Kiwi Kandy - WooCommerce E-commerce Platform

- Built and maintained complete WooCommerce e-commerce platform with custom product configurations, shopping cart functionality, and secure payment gateway integrations.
- Implemented Google Merchant Center integration for product feed management, enabling automated product listings and inventory synchronization across Google Shopping.
- Managed end-to-end order fulfillment workflows including automated shipping label generation, inventory tracking, and customer notification systems.

Work Experience

Fullstack Developer

August 2024 - Current

Rank / Web Merchants | Auckland CBD

- Architected and deployed production-ready AI chatbot systems using LangChain, OpenAI GPT-4, and RAG (Retrieval-Augmented Generation) with FAISS vector databases, enabling intelligent document-based question answering and automated lead capture with 95%+ accuracy.
- Built full-stack chatbot platforms with React frontends and Node.js/Express backends, implementing dynamic tool creation, session management, and SMTP email automation for real-time lead notifications.
- Developed custom LLM-powered APIs integrating OpenAI and Google Gemini, creating automation tools that reduced manual processing time by 70% and provided competitive advantages through intelligent workflow automation.
- Engineered PDF processing pipelines with text extraction, chunking, and embedding generation using OpenAI's text-embedding-ada-002 model, enabling semantic search across large document corpuses.
- Managed and optimised WooCommerce e-commerce operations including Google Merchant Center integration, product feed generation, inventory management, and order fulfillment workflows for multi-product online stores.
- Implemented secure authentication systems with session management, user roles, and API key generation for multi-tenant chatbot platforms.
- Developed custom WordPress plugins for WooCommerce, including event booking systems with date management, inventory tracking, and seat allocation logic, serving multiple client businesses.
- Built WooCommerce extensions with custom meta boxes, admin interfaces, AJAX functionality, and frontend integrations for complex product types and booking systems.
- Developed custom websites and landing pages with advanced JavaScript integrations including Google Maps API, payment gateways, and real-time data visualisation.
- Administered client LAMP stack environments, using SSH for server management, debugging complex issues, and optimising database queries for improved performance.

Business Development Consultant

March 2024 - August 2024

PENINSULA | Auckland CBD

- Enhanced verbal and written communication skills, improving client interactions and team collaboration.
- Translated complex technical concepts for non-technical clients, enhancing stakeholder communication.
- Operated in a metrics-driven environment, linking technical work to measurable business outcomes (KPIs).

Senior Repair Tech & Manager

July 2016 - Nov 2023

SMASHED IT / ILAB

- Managed and trained technical staff in diagnostics, repairs, and operational duties.
- Performed website management, SEO optimisation, and ad content creation to drive business growth.
- Authored process guides, training documents, and standard operating procedures (SOPs) for quality control.
- Diagnosed and resolved a wide range of complex hardware and software issues for clients.

Core Competencies & Skills

AI & Machine Learning

LangChain, LangGraph (Multi-Agent Systems), OpenAI API (GPT-4, GPT-4o, text-embedding-ada-002), Google Gemini API, RAG (Retrieval-Augmented Generation), FAISS & pgvector (Vector Databases), PDF Processing & Text Extraction, Web Scraping & Content Classification, LLM Prompt Engineering, Agent Tool Development, Chatbot Development, Semantic Search, OAuth 2.0 Integration

Front-End

React, TypeScript, JavaScript, Next.js, React Native, HTML5, CSS3, Tailwind CSS, Mantine UI, Figma, Photoshop, After Effects, Premier Pro

Back-End

Node.js, Express.js, Python, Flask, PHP, WordPress, WooCommerce Plugin Development, REST APIs, WebSockets, Session Management, Authentication & Authorisation, MySQL, PostgreSQL, DynamoDB, Sequelize ORM

DevOps & Cloud

Docker, Docker Compose, CI/CD, GitHub Actions, AWS (EC2, S3, RDS, Lambda), Google Cloud Platform (OAuth 2.0, Search Console API), Linux Server Administration, SSH, Apache, NGINX, Proxmox VE, Virtualisation, Redis (Session Management)

Automation & Business Platforms

n8n.io Workflow Automation, Web Scraping (Cheerio, JSDOM), RSS Feed Parsing, Streamlit, Nodemailer (SMTP), WooCommerce (E-commerce Management, Product Configuration, Payment Gateways), Google Merchant Center (Product Feed Management), Order Fulfillment Systems, Odoo, HubSpot, SEO Optimisation, Google Search Console Integration, Analytics & Tracking

Education

Bachelor of Science, Computer Science

Graduated Feb 2024

University of Auckland | Auckland, New Zealand